

RANJAN KUMAR VERMA

Spacecraft Operations Engineer | Scientist/Engineer 'SC', ISRO | TT&C & Payload Responsible Engineer
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PROFILE

Spacecraft engineer with a systems-level view of satellites in operation, built from a dual mandate at ISRO's Master Control Facility: **Responsible Engineer for the TT&C and Payload subsystems** (Spacecraft Payload & TT&C Analysis Division) and **spacecraft controller with 24×7 real-time commanding authority** over the GEO/GSO fleet (GSAT, IRNSS/NavIC, INSAT-3D series). Work spans cross-subsystem behaviour analysis, anomaly investigation, authoring of operational directives and contingency procedures, and design-review participation.

EXPERIENCE

Scientist/Engineer 'SC', Spacecraft Operations (SPTA Division)

Oct 2024 to Present

Master Control Facility (MCF), ISRO • Hassan, Karnataka

Systems Analysis & Responsible Engineer: Payload & TT&C

- **Health monitoring and signature-trend analysis** of communication and navigation payloads and TT&C subsystems across the GSAT and IRNSS (NavIC) fleets, including Rubidium Atomic Frequency Standard (RAFS) monitoring and periodic frequency-offset corrections.
- **Conduct cross-subsystem investigations**, correlating thermal behaviour with payload mechanism performance.
- **Author operational directives** and payload contingency recovery/reconfiguration procedures.
- **Define critical-alarm monitoring limits** for subsystem parameters and maintain the centralized anomaly database.
- **Manage transponder operations**: attenuation planning to user requirements, transponder allocation, overdrive and interference mitigation, and geolocation of interference sources.
- **Supported LEOP and In-Orbit Testing (IOT)** for a newly launched GEO communication satellite; presented subsystem findings at Spacecraft Operations Review Team (SORT) meetings.

Spacecraft Controller (24×7 rotational shifts)

- **Execute real-time and time-tagged commanding** across all subsystems: navigation coefficient uplinks, ranging per flight-dynamics requirements, reaction-wheel desaturation, and thruster testing prior to manoeuvres.
- **Support station-keeping (NSSK/EWSK)**, drift/eccentricity correction and eclipse-season operations, covering battery depth-of-discharge computation, reconditioning and cell balancing, plus lunar-shadow and meteorological payload (imager/sounder) operations.

Software Development

- **Built a GUI-based integrated health-check utility** for TT&C and payload monitoring, covering automated RAFS trend monitoring, long-term parameter plotting and payload event listing; adopted for division-wide health analysis.

EDUCATION

B.Tech, Electronics & Communication Engineering (Avionics), CGPA 7.80/10

2020 to 2024

Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram

PROJECTS & INTERSHIPS

Research Intern, MEMS Microfabrication (INUP), IIT Bombay

Mar to May 2024

- Fabricated a polymer (SU-8) MEMS piezoelectric energy harvester in IIT Bombay's cleanroom under the Indian Nanoelectronics Users Programme, realizing my IIST final-year design (simulated in CoventorWare), giving end-to-end design, simulation and fabrication exposure to how process constraints feed back into design.

Research Intern, SDR-Based FMCW Radar, IIST

May to Jul 2023

- Implemented an FMCW radar on ADALM-Pluto SDR in GNU Radio: chirp generation, beat-signal spectral analysis for multi-target range detection, extended to velocity estimation via range-Doppler processing.

EPS Team Member, Student Satellite Programme (SSPACE), IIST

Jan to May 2023

- Worked on the Electrical Power Subsystem for a student CubeSat, covering power budget analysis, solar-cell selection and sizing, and EPS board development.

SKILLS

Systems & operations: cross-subsystem behaviour analysis, anomaly investigation, operations procedures & directives, review-board participation (SORT), telemetry trend analysis, TT&C, payload & transponder operations, GEO/NavIC fleet operations

Programming & tools: Python (GUI utilities, ops automation), MATLAB, Linux, GNU Radio, ADALM-Pluto SDR, CoventorWare

Languages: English (professional), Hindi (native)

TRAINING, TALKS & ACHIEVEMENTS

- Presented "Starlink and OneWeb: Satellite-Based Internet Services" to the MCF Scientific Council (Jul 2025).
- **Spacecraft Operations Training Program (SOTP)**, MCF, Sep 2025.
- **ISRO Induction & Training Program (IITP-36)**, URSC, Nov to Dec 2025.
- **JEE Advanced 2020: All India Rank 4502.**